

MARAGE: Multi-Agent RAG-Enhanced System for Mitigating Hallucinations

Stephanie Yang & Aprille Xi & Jiacheng Wang
sy2@andrew.cmu.edu & ajxi@andrew.cmu.edu & jiachen8@andrew.cmu.edu
10-423/623 Generative AI Course Project

August 21, 2026

1 Introduction

LLMs frequently hallucinate, generating confident but factually incorrect responses even on high-stakes medical questions where errors have real consequences. While hallucination cannot be fully eliminated [Xu et al., 2025], multi-agent pipelines have shown promise in progressively reducing it. Gosmar and Dahl [2025] demonstrate this with a four-agent system achieving a cumulative 2,737% THS reduction, however, their reviewer agents do not consult external sources, limiting corrections to what the model already parametrically knows is wrong.

We introduce MARAGE (Multi-Agent RAG-Enhanced), a system that extends Gosmar and Dahl’s pipeline by replacing the second-level reviewer with a retrieval-augmented agent grounded in a local MedRAG corpus [Xiong et al., 2024]. A medical classifier first routes each prompt; if medical, a FAISS index over approximately 126k textbook passages is queried and the retrieved evidence is used to rewrite the response. If not medical, the original reviewer behavior is preserved. We evaluate both systems on 500 hallucination-inducing prompts (250 medical, 250 general) using a simplified THS metric and hand-labeled ground truth. Our results show that the MedRAG pipeline improves the within-pipeline 1→2 correction from 10.85% to 13.11% over the baseline, and reduces the human-labeled hallucination rate from 46% to 34%. These gains are concentrated at Agent 2, confirming that retrieval grounding strengthens the corrective step it was designed to improve.

2 Dataset / Task

Dataset. We generate 500 hallucination-inducing prompts following all of the 5 techniques listed in Gosmar and Dahl [2025]:

1. *Exploiting Knowledge Gaps and Underspecified Requests*, using obscure or ambiguous queries

about fictional entities

2. *Combining Fact and Fiction*, using confidently incorrect assertions or contradictory context to mislead the model
3. *Pressing Beyond the Model’s Limits*, requesting specific references or citations from non-existent sources
4. *Instructing the Model to be Creative or Speculative*, using role-playing or speculation prompts that blur fact and fiction
5. *Prompt Engineering and Instruction*, explicitly de-emphasizing factual accuracy or overriding the model’s expressed uncertainty.

An example of prompts generated for the knowledge gap criteria (technique 1) are as such:

”Could you explain the contributions of Loxoritean archaeologists to understanding ancient Zephyrilian hieroglyphics?” - General

”Describe the latest clinical trial for curing Xerostomia without saliva glands in detail.” - Medical

Medically-focused prompts are designed to test factual grounding against an authoritative medical corpus; in our case, they are the MedRAG textbooks. In total, the medical prompts will constitute 50% of the data. Further prompts were generated using Ollama with Qwen2.5 and manually reviewed to confirm if they were sufficiently ambiguous or likely to elicit hallucinated responses.

Task & Metric. Given this dataset, we define the following evaluation objective: our multi-agent pipeline must attempt to reduce the Total Hallucination Score (THS) of the final response given a hallucination-prone prompt. The THS is measured through our KPI Evaluator which is prompted to measure factuality and

helpfulness from a scale of 1 to 10. These are composited into the THS metric :

$$\text{THS}_n = \frac{\text{Factuality}_n + \text{Helpfulness}_n}{2} \quad (1)$$

A higher metric of THS indicates less hallucination and better factuality. This metric is an adjustment from the original Gosmar and Dahl [2025] paper which claimed four indicator metrics they also averaged. However, upon further inspection of their source code, this metric was found to be not used or implemented. Our metric was chosen to reflect most simply what hallucination means. To ground our metric, we have additionally hand-labeled 100 prompts (50 medical, 50 general) to compare against.

3 Related Work

Prior work explores several approaches to mitigating hallucinations. We consider many of the new techniques used as an alternative to modifying model weights through post-processing and assess our system. Previous research has mostly focused on retrieval-augmented generation (RAG) and reasoning methods like chain-of-thought (CoT) and tool-augmented reasoning [Li et al., 2025].

RAG. Retrieval-augmented generation for factual grounding is a general purpose framework combining a pre-trained dense retriever with a sequence-to-sequence generator over a non-parametric document index first introduced by Lewis et al. [2021]. Results indicate that this architecture produces responses that are more specific, diverse, and factually grounded than parametric-only baselines, which underlies the integration of our MedRAG agent [Xiong et al., 2024].

Agentic Reasoning. Gosmar and Dahl [2025] introduce the pipeline that we extend, which is a four-agent system where successive reviewer agents add disclaimers and reframe speculative content to achieve a cumulative THS reduction of 2737%. A key limitation is that reviewer agents operate entirely on parametric knowledge, thus only surfacing errors the model already recognizes. Wu et al. [2025] extend this further by integrating external tool-using agents directly into the LLM reasoning process, substantially improving factual reliability. Li et al. [2025] finds that combining RAG with agentic orchestration yields stronger hallucination reduction than any single technique in isolation, supporting our hybrid design.

Medical RAG. Our system targets medical prompts specifically because this domain combines high hallucination risk with high real-world consequence. Tran et al. [2025] directly inspire our secondary agent,

proposing a multi-step verification pipeline for medical chatbots that webcrawls authoritative sources. The retrieval backbone of our system draws on ?, which introduces the MedRAG toolkit compiling corpora from PubMed, medical textbooks, StatPearls, and Wikipedia, demonstrating RAG improves LLM accuracy by up to 18% over chain-of-thought prompting alone. In our implementation, we index the MedRAG textbook corpus using FAISS instead of GeoSI’s live web crawl.

Medical Routing Gate. Our system uses a zero-temperature `qwen2.5:3b-instruct` call to classify whether a prompt is medically related before routing it to the retrieval pathway. While our current classifier is prompt-based, Lee et al. [2019] demonstrate that domain-adaptive pre-training on biomedical corpora substantially outperforms general-domain models on medical text classification, suggesting a fine-tuned BioBERT gate as a natural future extension.

4 Methods

Our approach extends the multi-agent hallucination mitigation pipeline of Gosmar and Dahl by replacing the second-level reviewer with a retrieval-augmented variant grounded in a local MedRAG corpus, as illustrated in Figure 1.

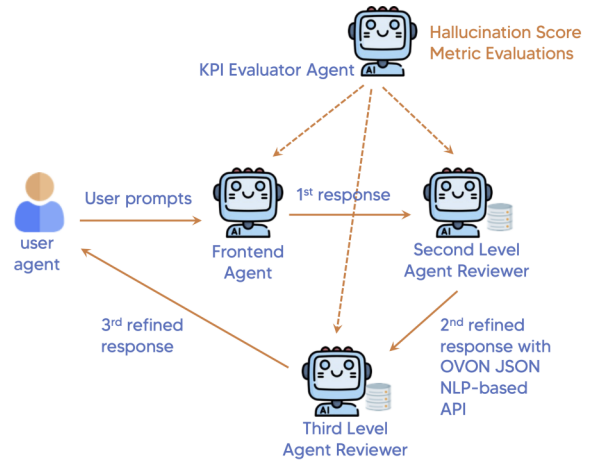


Figure 1: Adapted Figure from Gosmar and Dahl [2025].

Baseline System. The original pipeline consists of three sequential LLM agents followed by a KPI evaluator. The *FrontEndAgent* generates an initial response without constraints, intentionally allowing hallucinations to surface. The *SecondLevelReviewer* then revises this response, improving clarity and flagging speculative or unsupported claims. It produces a structured output consisting of (1) a rewritten response, (2) a short

hallucination risk summary, and (3) a detailed explanation of potential issues. The *ThirdLevelReviewer* further refines the output for style and factuality. Finally, the *KPI Evaluator* assigns scores and computes the Total Hallucination Score (THS).

We reimplement this pipeline as our baseline to establish a direct point of comparison reported by Gosmar and Dahl and establish a performance baseline.

Proposed System. We modify the second-stage reviewer by introducing retrieval-augmented grounding while keeping all other components unchanged. This design isolates the effect of retrieval on the primary correction stage.

The updated pipeline introduces two additional components at Agent 2: a medical classifier and a retrieval module, as shown in Figure 2.

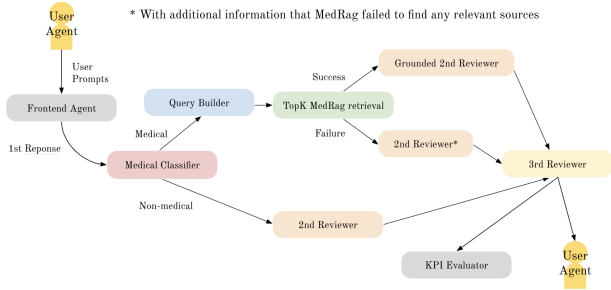


Figure 2: Proposed pipeline architecture with medical routing gate and MedRAG retrieval.

Step 1: Medical Classification. Each input prompt is first passed to a *MedicalClassifier*, implemented as a zero-temperature call to `gwen2.5:3b-instruct`. The classifier determines whether the query is medical in nature. Medical prompts are routed to the retrieval pipeline, while non-medical prompts follow the original reviewer path. This preserves backward compatibility with the baseline system.

Step 2: Query Construction. For medical prompts, a *MedRAGQueryBuilder* extracts 1-3 concise keyword phrases from the input. This step removes hallucination-inducing phrasing and improves retrieval precision by focusing on core semantic content.

Step 3: Retrieval. The extracted keywords are used to query a FAISS index built over approximately 126k passages from the MedRAG textbook corpus. Passages are embedded using `nomic-embed-text` and retrieved using cosine similarity. We select the top-6 passages above a threshold of 0.15.

If no passages meet the threshold, the system falls back to a retrieval-aware reviewer that explicitly signals uncertainty. This behavior is important, as failure to retrieve relevant information may indicate that the prompt contains fabricated or non-existent concepts.

Step 4: Grounded Generation. The retrieved passages are formatted with source labels and provided to the reviewer alongside the original prompt and initial response. The agent then generates a revised answer constrained to the retrieved evidence.

Output Format. All agents communicate using a lightweight JSON schema:

```

{
  "utterance": "<rewritten response>",
  "whisper_context": "<hallucination risk summary, max 20 words>",
  "whisper_value": "<detailed explanation of flagged claims, max 200 words>"
}

```

This structure preserves key communication signals while simplifying the original OVON format.

Summary. By combining multi-agent refinement with conditional retrieval, the system addresses two key failure modes: insufficient self-correction and reliance on ungrounded parametric knowledge. The modification is localized to the second-stage reviewer, enabling a controlled evaluation of retrieval augmentation.

5 Experiments

Hardware and Timing. All experiments were conducted locally on a CPU-only Mac using Ollama with `gwen2.5:3b-instruct`. The baseline pipeline processed 500 prompts in approximately 2 hours, while dataset generation required an additional hour. The MedRAG-enhanced pipeline introduced overhead from retrieval and query construction but completed within 3 hours total. No GPU acceleration was used.

Baseline Results. We first evaluate the reimplemented Gosmar and Dahl pipeline on all 500 prompts. As shown in Figure 3, the Total Hallucination Score (THS) improves by 13.43% from Agent 1 to Agent 3. Improvements are consistent across domains, with general and medical prompts achieving gains of 13.40% and 13.45%, respectively.

Most of the improvement occurs between Agent 1 and Agent 2, indicating that the second-stage reviewer performs the primary corrective role, while the third-stage reviewer contributes smaller refinements.

Medical prompts begin with a higher initial THS (6.50 vs. 5.97), suggesting that the model produces more confident or verbose responses in medical contexts. Human evaluation of a subset of 100 prompts yields a hallucination rate of 46%, with average factuality of 0.54 and helpfulness of 3.08.

Proposed System Results. We next evaluate the MedRAG-enhanced pipeline on the same dataset. The medical classifier routes 224 out of 500 prompts (44.8%) to the retrieval pathway. All routed prompts successfully retrieve at least one passage above the similarity threshold, with an average top score of 0.754.

Main Results. Under this metric, the MedRAG system consistently outperforms the baseline:

- Overall: 13.11% vs. 10.85%
- Medical prompts: 12.59% vs. 10.19%
- General prompts: 13.64% vs. 11.56%

These results indicate that retrieval grounding strengthens the corrective capacity of the second-stage reviewer.

Routing Analysis. Figure 4 provides a breakdown by routing group. Prompts correctly routed to MedRAG achieve a 13.50% improvement, compared to 12.79% for the original path. However, misclassification introduces limitations. General prompts incorrectly routed to MedRAG show negative downstream effects (6.16% from Agent 2 to 3) Medical prompts routed to the original path show reduced gains (8.01% overall improvement). This highlights the importance of accurate routing for system performance.

Human Evaluation. Human evaluation further supports the effectiveness of retrieval grounding. The MedRAG system reduces hallucination rate from 46% to 34%, while improving factuality from 0.54 to 0.66. Helpfulness remains largely unchanged. These results suggest that retrieval improves factual correctness beyond what is captured by the automatic THS metric, which may under-differentiate fine-grained improvements due to its reliance on a small LLM evaluator.

Summary. From Figure 3, we see that across all conditions, THS scores increase with each additional agent, consistent with Gosmar and Dahl’s original findings. Overall, the results demonstrate that retrieval augmentation improves the effectiveness of the second-stage reviewer, particularly for medical prompts. However, performance is sensitive to routing accuracy, and future improvements should focus on more robust classification and evaluation methods.

Condition	Subset	THS1	THS2	THS3	1→2 (%)	1→3 (%)
Baseline	All	6.23	6.91	7.07	10.85	13.43
Baseline	Medical	6.50	7.16	7.37	10.19	13.45
Baseline	General	5.97	6.66	6.77	11.56	13.40
MedRAG	All	6.09	6.89	6.97	13.11	14.37
MedRAG	Medical	6.16	6.94	7.06	12.59	14.50
MedRAG	General	6.03	6.85	6.88	13.64	14.24

Figure 3: Automatic KPI summary for the baseline and MedRAG pipelines by domain.

Group	Rows	THS1	THS2	THS3	1st→2nd (%)	2nd→3rd (%)	Avg Top Score
RAG	224	6.15	6.98	6.99	13.50	0.16	0.754
Orig	276	6.05	6.82	6.95	12.79	1.91	n/a
Gen+RAG	15	6.20	7.03	6.60	13.44	-6.16	0.694
Gen+Orig	235	6.01	6.84	6.90	13.65	0.96	n/a
Med+RAG	209	6.15	6.98	7.02	13.50	0.62	0.758
Med+Orig	41	6.24	6.74	7.24	8.01	7.41	n/a

Figure 4: MedRAG system THS results broken down by routing group.

Metric	Baseline	MedRAG
Hallucination Rate	46%	34%
Avg Factuality (0-1)	0.54	0.66
Avg Helpfulness (0-5)	3.08	3.20

Table 1: Hand-labeled evaluation on 50 responses for baseline and MedRAG systems.

6 Code Overview

The core of our proposed pipeline is `process_prompt` in `medrag_original.py` (lines 263-396), which is the unit of work dispatched to each thread in the parallel executor. The algorithm, like other parts of the codebase, was written first as pseudocode before completed by AI.

The function runs five sequential steps per prompt. First, the `FrontEndAgent` generates an unconstrained response (line 278), immediately followed by the `MedicalClassifier` deciding routing (lines 280-281). If medical, a `MedRAGQueryBuilder` extracts keywords (line 287), which are cleaned and passed to FAISS retrieval (line 289). A successful retrieval produces a grounded rewrite; a failed retrieval passes an explicit failure signal to a skepticism-primed fallback reviewer instead (lines 311-326). Non-medical prompts skip all of this and go directly to the standard `SecondLevelReviewer` (line 328). Both paths converge at the `ThirdLevelReviewer` and `KPI_Evaluator` (lines 340-352), which score all three agents simultaneously.

Two engineering details are worth noting. KPI parse failures are caught explicitly (lines 354-367) and logged to `error.txt` by prompt ID so they can be rerun individually without restarting the full pipeline. The result dictionary (lines 372-393) stores every intermediate value to create a full audit trail for every prompt in the output CSV.

For the reason of being the core to our methodology, and with the human detailing of all possible error fallbacks in our pipeline, we have chosen to highlight this function specifically.

7 Timeline, Human Work, & AI Use.

Time Spent. Table 2 summarizes the approximate hours spent on each stage of the project across all three team members.

Stage	Hours
Reading papers & documentation	9
Understanding & running existing code	7
Writing & modifying code	15
Running experiments & compiling results	10
Writing this document	6
Total	47

Table 2: Approximate hours spent per project stage.

Human Work. All core technical work was human-driven. System design decisions, pipeline architecture, dataset construction, and analysis and interpretation of results were all developed independently without AI assistance. The multi-agent pipeline was reimplemented from scratch, and all implementation logic was written by hand. All papers and resources were read in full by team members; AI summaries were used only as an initial filter to prioritize reading order, with all interpretations and decisions made from the original sources.

AI Use. We used AI as a writing assistant for grammar checking and clarity edits on this report. It was also used to help sift through related work at a high level, though all cited papers were read in full by at least one team member before being included. For implementation, AI tools were used for fine-grain code writing with human pseudocode/outlines.

All AI-generated code suggestions were manually reviewed, tested against expected outputs, and cross-checked against library documentation before being incorporated. No AI tool was used to generate prompts, label data, or interpret experimental results.

8 Research Log

Our original proposal began with a clear plan to replicate Gosmar and Dahl’s multi-agent hallucination mitigation pipeline and extend it with GeoSI, a retrieval agent for medical prompts, and then compare the two systems on a dataset we constructed ourselves. In practice, almost every component of that plan required significant changes.

Replicating the Baseline. The Gosmar and Dahl paper reports a 2,737% cumulative THS reduction, which is a striking headline result. When we examined their GitHub code, we found the pipeline was not runnable end-to-end. The KPI evaluator prompt asked for FCD,

FGR, FDF, and ECS metrics but specified no exact JSON schema or any description of the meaning of these 4 metrics. Thus, we had to reimplement the pipeline from scratch, building a checkpoint system that atomically writes partial results every N prompts, and parallelizing prompt processing.

The THS Metric. The original formula is:

$$\text{THS} = \frac{w_1 \text{FCD} - (w_2 \text{FGR} + w_3 \text{FDF} + w_4 \text{ECS})}{N_A} \quad (2)$$

where FCD counts factual claims per 100 words, FGR counts references to "evidence" or "records," FDF counts mentions of "fiction" or "myth," and ECS is a binary score for explicit fictional framing. As previously mentioned, the metric lacked clear implementation and justification. We therefore replaced it with a simpler rubric as described. Because our revised THS is still an LLM-judged metric, we added a hand-labeling to get human ground truth on hallucination rate and helpfulness.

GeoSI was not Integrable. Our midway proposal planned to replace the second-level reviewer with an agent that crawls authoritative medical sources (WHO, PubMed) using the GeoSI system. When we attempted integration, we found that GeoSI’s retrieval relies on live Google Search to surface and crawl pages. This was a hard blocker; we were unable to run searches on 500 prompts and quickly got bot-restricted. We pivoted to the MedRAG corpus instead and built a local FAISS index over the textbooks subset served by Ollama. This gave us a fully offline, reproducible retrieval system.

9 Conclusion

We presented MARAGE, a multi-agent hallucination mitigation system that augments the second-stage reviewer with domain-conditioned retrieval. Across 500 prompts, the retrieval-augmented pipeline consistently improves the critical Agent 1 → 2 correction step, indicating stronger intermediate revision. Human evaluation further supports these gains, reducing hallucination rate from 46% to 34% while improving factuality without degrading helpfulness.

Several limitations remain. The KPI evaluator is a relatively small model and introduces instability in automatic metrics; using a stronger evaluator would improve reliability. Human evaluation is limited to a 50-sample subset and should be expanded to reduce noise. In addition, routing errors degrade performance when prompts are misclassified, highlighting the need for a more robust, fine-tuned classifier.

Overall, these results demonstrate that integrating retrieval into multi-agent pipelines is an effective and

modular strategy for reducing hallucination, particularly in high-stakes domains such as medicine.

10 Thought-Experiment on Compute

Actual Compute Use. All experiments were run locally on a personal Mac, using CPU only via Olama. We estimate approximately 2 hours of runtime for the baseline pipeline across 500 prompts, and 1 hour for prompt dataset generation, totaling roughly 3 CPU hours. Since consumer Mac CPUs have no direct cloud equivalent pricing, the closest comparable cloud instance would be a T4 GPU at \$0.53/hr, which would place our compute at an estimated \$1.59 in cloud-equivalent cost. In practice, local CPU inference with `qwen2.5:3b-instruct` was significantly slower than GPU inference would have been, and the small model size was chosen specifically to remain feasible under these constraints.

With \$1,000 of Cloud GPU Credits. With additional compute, we would make three changes. First, we would scale up the model size. Rather than `qwen2.5:3b-instruct`, we would run a 70B-parameter model such as Llama-3-70B across all pipeline agents. At the L40S rate of \$1.86/hr, \$1,000 provides approximately 537 GPU hours. Second, we would expand our prompt dataset substantially, targeting 2,000-3,000 prompts rather than 500, with a richer breakdown of medical subcategories to better stress-test the GeoSI retrieval component. Third, we would run a proper hyperparameter search for the BioBERT routing gate, experimenting with learning rate, batch size, and training set size, which was not feasible locally. Together these changes would produce more statistically robust results and a stronger test of whether retrieval augmentation generalizes across model scales.

References

Diego Gosmar and Deborah A. Dahl. Hallucination mitigation using agentic ai natural language-based frameworks, 2025. URL <https://arxiv.org/abs/2501.13946>.

Jinhyuk Lee, Wonjin Yoon, Sungdong Kim, Donghyeon Kim, Sunkyu Kim, Chan Ho So, and Jaewoo Kang. Biobert: a pre-trained biomedical language representation model for biomedical text mining. *Bioinformatics*, 36(4):1234–1240, September 2019. ISSN 1367-4811. doi: 10.1093/bioinformatics/btz682. URL <http://dx.doi.org/10.1093/bioinformatics/btz682>.

Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich K ttler, Mike Lewis, Wen tau Yih, Tim Rockt schel, Sebastian Riedel, and Douwe Kiela. Retrieval-augmented generation for knowledge-intensive nlp tasks, 2021. URL <https://arxiv.org/abs/2005.11401>.

Yihan Li, Xiyuan Fu, Ghanshyam Verma, Paul Buitelaar, and Mingming Liu. Mitigating hallucination in large language models (llms): An application-oriented survey on rag, reasoning, and agentic systems, 2025. URL <https://arxiv.org/abs/2510.24476>.

Trong-Nghia Tran et al. Enhancing medical chatbot reliability: A multi-step verification approach to prevent hallucinations. In *Proceedings of the 2nd Workshop on Security-Centric Strategies for Combating Information Disorder*, pages 1–12, 2025. doi: 10.1145/3709020.3734832.

Junde Wu, Jiayuan Zhu, Yuyuan Liu, Min Xu, and Yueming Jin. Agentic reasoning: A streamlined framework for enhancing llm reasoning with agentic tools, 2025. URL <https://arxiv.org/abs/2502.04644>.

Guangzhi Xiong, Qiao Jin, Zhiyong Lu, and Aidong Zhang. Benchmarking retrieval-augmented generation for medicine, 2024. URL <https://arxiv.org/abs/2402.13178>.

Ziwei Xu, Sanjay Jain, and Mohan Kankanhalli. Hallucination is inevitable: An innate limitation of large language models, 2025. URL <https://arxiv.org/abs/2401.11817>.